EXPONENTIAL SMOOTHING AND UIC ENROLLMENTS FORECASTING

A brief introduction to Exponential Smoothing

Exponential smoothing is a method of forecasting that uses a weighted average of past observations to predict future values. Unlike the simple moving average, where past observations are weighted equally, exponential smoothing assigns exponentially decreasing weights over time. Essentially, more recent observations have a higher weight than older ones. This technique is commonly used for analyzing time-series data. The simplest form of exponential smoothing is given by the formulas:

$$s_{(t)} = \alpha x_{(t)} + (1-\alpha)s_{t-1}$$

where:

- $s_{(t)}$ is the smoothed value at time t,
- $x_{(t)}$ is the observed value at time t,
- s_{t-1} is the previous smoothed statistic, and
- α is the smoothing parameter between 0 and 1.

The smoothing parameter α controls the weight given to the current observation and the previous forecast.

A high value of α gives more weight to the current observation, while a low value of α gives more weight to the previous forecast.

Holt's linear exponential smoothing, also known as double exponential smoothing, is used to forecast time series data that has a linear trend but no seasonal pattern. This method uses two smoothing parameters: α for the level (the intercept) and β for the trend.

The formulas for double exponential smoothing are:

$$s_t = \alpha x_t + (1 - \alpha)(s_{t-1} + b_{t-1})$$

$$b_t = \beta(s_t - s_{t-1}) + (1 - \beta)b_{t-1}$$

where:

- b_t is the slope and best estimate of the trend at time t,
- α is the smoothing parameter of data ($0 < \alpha < 1$), and
- β is the smoothing parameter for the trend ($0 < \beta < 1$).

Holt's method is more accurate than SES for time series data with a trend, but it doesn't work well for time series data with a seasonal component.

Need for Exponential Smoothing to forecast UIC Enrollments

The UIC Enrollments Forecasting team has utilized the ARIMA model to procure a 10-year forecast divided between two types of Student Codes: Continuing and Non-Continuing Students. The assumption behind this choice of division is that the continuing students enrollment is affected by a different set of parameters than the New first-year and other (Non-Continuing) students. For each of the above two groups, a new ARIMA model was run for each combination of Student Level (Undergraduate, Graduate, and Professional), College (COLL_CD in EDW), and Term (Fall, Spring, Summer) to create a 10-year forecast for each of the above combinations. In essence, if there are 3 colleges, then the number of ARIMA models generated was 27. Each of the model results were then collated into a single document to produce the 10-year ARIMA forecast output file.

This document deals with the results of the Non-Continuing Students Forecast only.

However, ARIMA model was not well-suited for certain combinations. Below is a list of combinations where ARIMA was unable to produce a reliable forecast:

- Undergraduate-Business Administration-Spring
- Undergraduate-Applied Health-Summer
- Graduate-Business Administration-Fall
- Graduate-Medicine-Fall
- Graduate-Dentistry-Spring
- Graduate-Pharmacy-Spring
- Graduate-Liberal Arts & Sciences-Summer
- Graduate-Medicine-Summer
- Professional-Applied Health-Spring
- Professional-Dentistry-Spring
- Professional-Nursing-Spring
- Professional-Dentistry-Summer

To handle these specific cases, during this iteration, we implemented an ad-hoc Exponential Smoothing model.

The next portion of this document gives a high-level explanation of the steps involved in the Exponential Smoothing Script.

Exponential Smoothing Model for UIC Enrollment Forecasts

As concluded in the first section of this document, we chose the Double Exponential Smoothing or Holt's Exponential Smoothing to make a forecast for the specific combinations listed in the second section of this document. The reason we decided to employ Holt's Exponential Smoothing is that: By dividing the enrollments dataset into Fall, Spring and Summer and running a separate model on these subsets, we essentially removed seasonality from the original dataset. Since there's an underlying trend in the subset with no seasonality, Holt's variation of Exponential Smoothing suits well for our use case.

Exponential Smoothing has been implemented using an ad-hoc Python script for forecasting time series data. It includes importing necessary libraries, defining a range of alpha and beta values for model optimization, loading and preparing data, splitting the data into training and validation sets, initializing variables for storing the best parameters, iterating over alpha and beta values to find the optimal model, fitting the best model to the entire dataset, making forecasts, calculating errors, and printing the results. The Python code[2] can be referred to in the Intermediate Files subfolder within the 10-year Projections Box Folder[1].

The below bullet points explain the Python code at a high-level and the Python file also includes explanatory comments for better understanding

1. The range of alpha (level) and beta (trend) values for model optimization (0.05 to 1 with a step of 0.05)
2. Load the enrollment data for various academic programs and terms into NumPy Arrays. Each array contains the number of enrollments for consecutive terms.
3. Split the data into training and validation sets. The training set contains 80% of the data; the validation set contains the remaining 20%.
4. Initialize variables to store the best alpha, beta, and RMSE (Root Mean Square Error) values.
5. The best values will be determined by iterating over the defined ranges and finding the lowest RMSE.
6. Iterate over all combinations of alpha and beta values. For each combination of alpha and beta value:
    a. Fit the Exponential Smoothing model to the training data with the current alpha and beta.
    b. Forecast the validation set using the fitted model.
    c. Calculate the RMSE for the forecast and compare it to the best RMSE found so far.
    d. If the current RMSE is lower, update the best alpha, beta, and RMSE values.
7. After finding the best alpha and beta values, fit the Exponential Smoothing model to the entire dataset.
8. Forecast the next 10 periods using the best-fitted model.
9. Calculate the MAE (Mean Absolute Error) for the best model using the validation set.
10. Determine the forecast range by subtracting and adding the MAE to the point forecast.
11. Print the best alpha, beta, RMSE, MAE, point forecast, and forecast range.

References:

1. 10-Year Projections Box Folder: https://uofi.box.com/s/hsbz3k6vm2klq4gcpvfwck4xfftl0h3p
2. Python Code performing Exponential Smoothing on UIC Non-Continuing Enrollments: https://uofi.box.com/s/z2vtfuwwtua95lqjvdayp87xckipeux8